

# Battery Supply Chain Risk System

LLM Multi-Agent Pipeline · NAATBatt Data (March 2026)

Enter a news article or supply chain query:

Major strike at Glencore cobalt mines in the Democratic Republic of Congo entered its third week Monday, with workers demanding higher wages amid record cobalt prices. The walkout has halted production at facilities supplying roughly 8% of global cobalt output, raising concerns among battery manufacturers in North America who rely on DRC-sourced material for NMC cathodes.

Sample inputs

S1 — Cobalt mine strike (D... ▾

Load sample

 Analyze Risk

 Analyse abgeschlossen

Severity

 3 / 5

Risk Type

Supply Disrup...

Material

Cobalt

Region

Africa/DRC

## Capacity Impact

Affected — NA Upstream

95.1%

Alternative — NA Upstream

4.9%

Affected — Global (USGS)

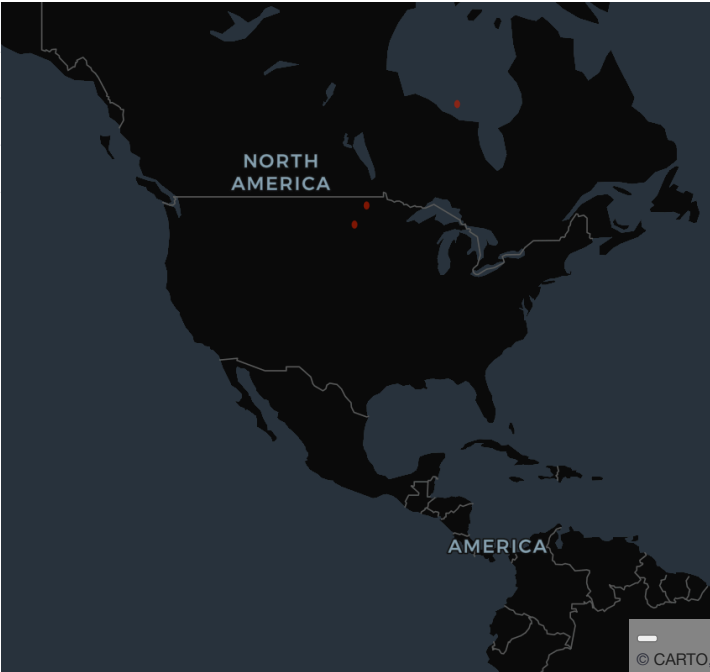
2.65%

Alternative — Global

0.14%

## Top 3 High-Risk Facilities

| Rank | Company          | Segment  | Location   | Score      |
|------|------------------|----------|------------|------------|
| 1    | Vale Canada Ltd. | Upstream | NL, Canada | 26.1 / 100 |
| 2    | Vale Canada Ltd. | Upstream | ON, Canada | 23.5 / 100 |
| 3    | Glencore         | Upstream | QC, Canada | 22.5 / 100 |



Risk Report

Risk Event

A major strike at Glencore cobalt mines in the Democratic Republic of Congo (DRC) has halted production, affecting 8% of global cobalt output. This disruption is significant for battery manufacturers relying on DRC-sourced material for NMC cathodes, particularly in North America. The strike's duration and impact on global cobalt supply will likely persist for at least 1-3 months. Source: NAATBatt facility data + LLM analysis.

Affected Material & Supply Chain Tier

Cobalt, upstream origin tier, supply disruption risk type.

Capacity Impact (North America)

- Affected NA capacity: 95.1%
- Alternative NA capacity: 4.9%
- Affected global capacity (USGS basis): 2.65%

Top 3 High-Risk Facilities

1. Vale Canada Ltd. — Voisey Bay, NL, Canada (Upstream, RiskScore: 26.1/100)
- This facility is high risk due to its significant capacity share in North America's upstream cobalt supply. Recommended action: Develop contingency plans with alternative suppliers to mitigate potential shortages.
2. Vale Canada Ltd. — Sudbury Basin - Copper Cliff Complex, ON, Canada (Upstream, RiskScore: 23.47/100)
- This facility is high risk due to its substantial capacity share in North America's upstream cobalt supply. Recommended action: Diversify suppliers and maintain open communication with key stakeholders to ensure timely information exchange.
3. Glencore — Raglan Mine, QC, Canada (Upstream, RiskScore: 22.52/100)
- This facility is high risk due to its significant capacity share in North America's upstream cobalt supply. Recommended action: Develop a risk management strategy to mitigate potential supply chain disruptions, including identifying alternative suppliers and negotiating long-term contracts.

Overall Risk Assessment

Severity 3/5 — A significant supply disruption in the DRC has halted 8% of global cobalt output, affecting North America's upstream capacity by 95.1%. The system's resilience is compromised due to limited alternative capacity, with only 4.9% available in North America.

▼ Pipeline details

Validation

 Issues found

Iterations

2

Validation issues:

- Risk type and severity do not match the described event (supply\_disruption, severity 3/5, but report mentions a major strike with significant impact)
- Report mentions Glencore cobalt mines in the DRC, but the affected region is listed as Africa/DRC in the context
- Justification for Vale Canada Ltd. (Voisey Bay) is missing
- Recommendation for Vale Canada Ltd. (Sudbury Basin - Copper Cliff Complex) is vague

- Report mentions Glencore Raglan Mine, but the context only lists Glencore as a facility, not a specific mine

Raw pipeline state (debug)

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{
  "raw_input" :
    "Major strike at Glencore cobalt mines in the Democratic Republic of Congo entered its third week Monday, with workers demanding higher wages amid record cobalt prices. The walkout has halted production at facilities supplying roughly 8% of global cobalt output, raising concerns among battery manufacturers in North America who rely on DRC-sourced material for NMC cathodes."
  "iteration" : 2
  "relevant" : true
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  "material" : "cobalt"
  "region" : "Africa/DRC"
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    2 : "cobalt"
    3 : "Democratic Republic of Congo"
    4 : "wages"
    5 : "prices"
    6 : "production"
    7 : "battery manufacturers"
  ]
  "filtered_text" :
    "Major strike at Glencore cobalt mines in the Democratic Republic of Congo entered its third week Monday, with workers demanding higher wages amid record cobalt prices. The walkout has halted production at facilities supplying roughly 8% of global cobalt output, raising concerns among battery manufacturers in North America who rely on DRC-sourced material for NMC cathodes."
  "severity" : 3
  "risk_type" : "supply_disruption"
  "affected_material" : "cobalt"
  "affected_region" : "Africa/DRC"
  "origin_tier" : "Upstream"
  "reason" :
    "A major strike at Glencore cobalt mines in the DRC has halted production, affecting 8% of global cobalt output. This disruption is significant for battery manufacturers relying on DRC-sourced material for NMC cathodes, particularly in North America. The strike's duration and impact on global cobalt supply will likely persist for at least 1-3 months."
  "betroffene_kapazitaet_pct" : 95.1
  "alternative_kapazitaet_pct" : 4.9
}
```

```
"risk_report" :
### Risk Event
A major strike at Glencore cobalt mines in the Democratic Republic of Congo (DRC) has halted production,
affecting 8% of global cobalt output. This disruption is significant for battery manufacturers relying on DRC-
sourced material for NMC cathodes, particularly in North America. The strike's duration and impact on global
cobalt supply will likely persist for at least 1-3 months. Source: NAATBatt facility data + LLM analysis.

## Affected Material & Supply Chain Tier
Cobalt, upstream origin tier, supply disruption risk type.

## Capacity Impact (North America)
- Affected NA capacity: 95.1%
- Alternative NA capacity: 4.9%
- Affected global capacity (USGS basis): 2.65%

## Top 3 High-Risk Facilities
1. **Vale Canada Ltd. – Voisey Bay, NL, Canada** (Upstream, RiskScore: 26.1/100)
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Recommended action: Develop contingency plans with alternative suppliers to mitigate potential shortages.
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North America's upstream capacity by 95.1%. The system's resilience is compromised due to limited alternative
capacity, with only 4.9% available in North America."
{
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      "segment": "Upstream"
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"issues" : [
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  1 :
    "Report mentions Glencore cobalt mines in the DRC, but the affected region is listed as Africa/DRC in the context"
  2 : "Justification for Vale Canada Ltd. (Voisey Bay) is missing"
  3 : "Recommendation for Vale Canada Ltd. (Sudbury Basin - Copper Cliff Complex) is vague"
  4 :
    "Report mentions Glencore Raglan Mine, but the context only lists Glencore as a facility, not a specific mine"
]
}
```